

Trần Mạnh Duy

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PROFILE

Go backend engineer with 3 years of experience building high-throughput, event-driven microservices for crypto trading platforms — Kafka stream processing, gRPC services, and correctness-sensitive fintech workloads. Owns services end to end in Go, from re-architecting inherited pipelines to diagnosing and remediating production incidents at scale.

SKILLS

Languages: Go (primary), Rust, C++

Backend: gRPC (protobuf), REST, WebSocket, hexagonal & event-driven microservices

Data & Streaming: Apache Kafka, Redis, PostgreSQL, MongoDB

Infrastructure: Docker, Kubernetes, ArgoCD, GitHub Actions, k6

EXPERIENCE

Sotatek

Backend Engineer

Hanoi, Vietnam

Jul 2023 – Present

RaidenX — Multi-chain DEX aggregator & trading-data platform

- Re-architected the **insight** trading-data service from an inherited single-chain TypeScript prototype into event-driven Go, then extended it across 5+ chains (Sui, BSC, Base, Solana, Monad) — driving the migration largely solo before the team scaled to 8–10 engineers.
- Owned the PnL/ROI pipeline: batched Kafka consumers aggregate balance-change events per wallet/token across ~20–25 partitions, recompute PnL/ROI, and upsert to PostgreSQL with in-batch dedup and duplicate-key retry (at-least-once, manual offset commit, in-consumer per-user ordering).
- Diagnosed and remediated a Kafka consumer-lag incident (~4M-message backlog in hours) by redesigning ingestion from per-message to batched processing — cutting steady-state lag from millions to the low hundreds and raising sustained throughput.
- Built a real-time price-alert service (MongoDB) with an in-memory threshold-crossing state machine, Redis pub/sub for cross-pod coherence, and dual Socket.IO + Kafka delivery; plus a trending-token engine using Redis sorted sets across 5m/1h/6h/1d windows.

Technology and related tools: Go, Apache Kafka, PostgreSQL, MongoDB, Redis, gRPC, Kubernetes

VDAX — Binance-style centralized exchange

- Built the referral reward/commission engine: multi-tier commission accrual on trades and settlements, deduplicated by a unique (user, transaction) constraint with decimal-precise money handling.
- Owned the notification pipeline via Kafka background workers — RabbitMQ transactional emails, Redis-backed Socket.IO real-time updates, and Firebase push — with idempotent delivery across deposit/withdrawal settlement events.
- Contributed to the accounts service: user lifecycle, Sumsu KYC, and multi-chain deposits/withdrawals with TOTP 2FA over PostgreSQL.

Technology and related tools: Go, Apache Kafka, PostgreSQL, Redis, RabbitMQ, Firebase

Kotoba Press — Japanese-learning platform

- Architected a hexagonal Go backend (Gin, MongoDB) with ports-and-adapters layering, domain error handling, **slog** logging, Google OAuth2 + JWT, and composition-root DI across API, importer, and indexer binaries.
- Integrated a custom C++17 BM25 search engine over gRPC/protobuf (client-streaming bulk indexing, gRPC-first search with MongoDB regex fallback), sustaining ~15K docs/sec indexing and ~0.04 ms median query latency.
- Load-tested the Go API with k6: 19K req/s on health and 4.4K req/s on vocabulary queries (p99 < 32 ms), stable to 800 concurrent users.

Technology and related tools: Go, Gin, gRPC, MongoDB, C++17, k6, Docker

EDUCATION

University of Engineering and Technology, VNU (UET-VNU)

B.Sc. Information Technology

Hanoi, Vietnam

Jul 2026